

Ms. Ref. No.: **KNOSYS-D-26-21490**

Title: *Effective Training Pressure Gates Recursive Knowledge Degradation in LLMs: A Multi-Axis Dose-Response Study*

Journal: *Knowledge-Based Systems* (Elsevier)

To: Dr. Hang Yu, Senior Editor

Re: Response to Reviewers — Revision Round 1

Date: 5 October 2026

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Dear Editor and Reviewers,

We thank you for your careful and constructive evaluation of our manuscript. Your comments prompted substantive revisions to the experimental account, statistical reporting, notation, and scope of our conclusions. We respond to each comment below and indicate the corresponding changes in the manuscript.

The most consequential correction concerns the original synthetic-exposure comparison. During revision, we identified that its baseline and intervention conditions had been evaluated through different pipeline paths, including differences in the construction of the baseline-correct question set ( $K_0$ ). Consequently, we no longer rely on that comparison as evidence for the previously reported intervention effect. The revised manuscript instead presents paired dose-response experiments conducted under consistent pipelines. On Qwen, reducing synthetic exposure attenuated the Generation 5–10 retention decline at the 50% reduction level. On Gemma 3, the corresponding intervention improved Generation 10 retention but did not attenuate the decline over that interval. We have revised the interpretation to preserve this distinction rather than proposing a common mechanism that the experiments do not establish.

We have also clarified the status of effective training pressure (ETP). It is now presented as an operational framework for organizing experimental conditions and their observed diagnostics, not as a validated universal scalar predictor or an architecture-independent threshold. Additional revisions distinguish controlled comparisons from descriptive contrasts, report the available seed-level uncertainty, define the notation used in the equations and figures, and state the limits of the factual-retention evaluation.

The responses that follow identify the action taken for each concern and direct the reader to the relevant part of the revised manuscript. Revised manuscript passages quoted in this letter appear in [blue](#).

Respectfully submitted on behalf of all authors,

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Corresponding Author

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## Conventions used in this response

- Reviewer comments are reproduced *in green italics*, verbatim, including the table and section numbers of the *original* submission.

- Text newly added to the manuscript appears “in blue”. Where a passage from the revised manuscript is reproduced in this letter, it is paraphrased or summarized; the relevant section is cited so that the exact wording can be located in the revision.
- Quoted manuscript passages are indicative wording unless marked [verbatim]; section references locate the exact text.
- Locations are given as section and page of the revised manuscript. Where the revision renumbered a table or section, both numbers are given at first mention, e.g. Table 7 of the original submission (now Table 9).
- All numerical values quoted here are generated from the per-seed results by `scripts/analysis/make_all.py` (Zenodo v1.0.4, <https://doi.org/10.5281/zenodo.23185146>) and are reproduced verbatim from its output.

## Experiment identifiers used in this response

The following identifiers appear in the revised manuscript and in this response. Table 3 of the revised manuscript maps the control experiments to the sources of variation they address.

*Table 3 from the revised manuscript — Controls and confounders*

| Source                      | Potential confound                                                        | Control                                                                                                                                                                 |
|-----------------------------|---------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Quantization (NF4 vs bf16)  | Numerical noise                                                           | G3: QLoRA bf16 vs NF4, matched seeds                                                                                                                                    |
| Training data order         | Shuffling artefacts                                                       | G4: 3 shuffle seeds at $r = 256$                                                                                                                                        |
| Optimizer state             | Accumulated momentum                                                      | Held constant: fresh optimizer every generation                                                                                                                         |
| Synthetic data seed         | Stochastic variability                                                    | Reported per seed; headline 3–5 seeds                                                                                                                                   |
| Backbone architecture       | Thresholds may not generalize                                             | Replicated on Gemma 3 ( $N = 5$ ) and Gemma 4 E2B ( $N = 1$ )                                                                                                           |
| Pipeline version            | Different code paths could introduce discrepancies                        | Each experiment uses a dedicated script; per-experiment script names documented in the repository README; per-execution commit traceability not systematically recorded |
| Evaluation scoring leniency | Bidirectional substring criterion may inflate retention at high verbosity | Noted per condition                                                                                                                                                     |

All 7 rows shown; complete formatted table in the revised manuscript (Section 3.9).

| ID    | Experiment                                        | Backbone      | Seeds                      | Role                                                                                                                                                           |
|-------|---------------------------------------------------|---------------|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| G1    | QLoRA rank sweep, NF4                             | Qwen 2.5 1.5B | 3 at $r=256$               | Primary dose-response                                                                                                                                          |
| G2    | Pre-registered exposure dose-response, $r=10$     | Gemma 3 1B IT | 3 per dose                 | Cross-backbone replication                                                                                                                                     |
| G2b   | Steps-matched control on Gemma 3 $r = 10$         | Gemma 3 1B IT | 3                          | Paired control for G2; inconclusive                                                                                                                            |
| G3    | Precision control: bf16                           | Qwen 2.5 1.5B | 3 at $r=16$ ; 3 at $r=256$ | Identical regime classifications under bf16; full causal conclusion pending per-item audit                                                                     |
| G4    | Data-order control: three new shuffle seeds       | Qwen 2.5 1.5B | 3 at $r=256$               | Directionally consistent with rank effect; tests data-order dependence (non-significant; seed and data order change simultaneously, neither variable isolated) |
| G5    | Pre-registered prospective rank $\times$ LR probe | Qwen 2.5 1.5B | 1 per cell                 | Exploratory ordering test                                                                                                                                      |
| B7    | Pre-registered exposure dose-response, $r=256$    | Qwen 2.5 1.5B | 5 per dose                 | Primary exposure evidence                                                                                                                                      |
| C1–C5 | Exploratory intervention pilot (archived)         | Qwen 2.5 1.5B | 1                          | Superseded by B7; excluded from all estimates                                                                                                                  |

## Summary of changes

Forty review points were raised: three by the Editor, five by Reviewer 1, and thirty-two by Reviewer 2. Each is answered individually below. Points marked PART are addressed in part, with the residual limitation stated explicitly in Section 6. The table maps every point to the revision made and its location.

| Point         | Issue raised      | Revision made                                                        | Location           |
|---------------|-------------------|----------------------------------------------------------------------|--------------------|
| <b>Editor</b> |                   |                                                                      |                    |
| ED-1          | Language editing  | Complete language revision; terminology standardized throughout      | Throughout         |
| ED-2          | Submission format | Seven documents in Word; figures at 300 dpi (non-PDF); .bib included | Submission package |
| ED-3          | List of changes   | This table plus an individual response to each point                 | This document      |

| Point                                                                                                                                                                                            | Issue raised                                                  | Revision made                                                                                                                                                                                            | Location                         |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| <b>Reviewer 1</b> ( <i>general assessment — outside the 40 counted points</i> )                                                                                                                  |                                                               |                                                                                                                                                                                                          |                                  |
| <i>Not a transferable predictor; single dataset; single-seed analyses; one exposure condition. → Framework reframed as operational; scope and seed policy stated per claim. [§3.5, §3.8, §6]</i> |                                                               |                                                                                                                                                                                                          |                                  |
| R1-1                                                                                                                                                                                             | Prospective calibration PART                                  | Five-step calibration protocol specified; held-out validation declared not performed                                                                                                                     | §3.5, §6                         |
| R1-2                                                                                                                                                                                             | Confound separation                                           | Controls G3 (precision) and G4 (data order); fresh optimizer per generation; design/observed split                                                                                                       | §3.9, Tab. 3, §4.6               |
| R1-3                                                                                                                                                                                             | External validity PART                                        | Scoring sensitivity documented as limitation; scope stated                                                                                                                                               | §3.2, §6                         |
| R1-4                                                                                                                                                                                             | Exposure: Qwen versus Gemma 3                                 | Pipeline inconsistency disclosed; two pre-registered dose–response experiments replace the pilot                                                                                                         | §3.7, §4.5                       |
| R1-5                                                                                                                                                                                             | Reproducibility                                               | Per-seed results, configs, evaluation protocol, K0 identifiers and question partition deposited (Zenodo v1.0.3); analysis and audit scripts on the GitHub HEAD, to be deposited in the next release; ... | §3.3, App. A                     |
| <b>Reviewer 2</b> ( <i>general assessment — outside the 40 counted points</i> )                                                                                                                  |                                                               |                                                                                                                                                                                                          |                                  |
| <i>Framework conceptual; Gemma 4 uncited; limited scale. → Contribution restated as design and evidence; Gemma 4 cited and disambiguated; scale stated as a limitation. [§1, §3.3, §6]</i>       |                                                               |                                                                                                                                                                                                          |                                  |
| R2-1.1                                                                                                                                                                                           | ETP in Fig. 1 before definition                               | Fig. 1 carries full name; ETP introduced in §1                                                                                                                                                           | §1, Fig. 1                       |
| R2-1.2                                                                                                                                                                                           | MTLD without expansion                                        | Expanded as “Measure of Textual Lexical Diversity”                                                                                                                                                       | §3.6                             |
| R2-1.3                                                                                                                                                                                           | KL, JS, coverage in Fig. 1 without definition                 | Removed from Fig. 1; replaced with Distinct- $n$ , MTLD, stopword ratio                                                                                                                                  | Fig. 1                           |
| R2-1.4                                                                                                                                                                                           | bf16 without expanded form                                    | Expanded as “bfloat16 (bf16)” at first use                                                                                                                                                               | §3                               |
| R2-1.5                                                                                                                                                                                           | SVD without expansion                                         | Expanded as “singular value decomposition”                                                                                                                                                               | §2.4                             |
| R2-1.6                                                                                                                                                                                           | TCE without expansion                                         | Expanded as “Truncated Cross-Entropy”                                                                                                                                                                    | §2.3                             |
| R2-1.7                                                                                                                                                                                           | QLoRA/LoRA inconsistent expansion                             | Both expanded at first use in §1                                                                                                                                                                         | §1; §2.4                         |
| R2-1.8                                                                                                                                                                                           | “Homeo.” in Table 2 undefined                                 | “Homeostatic” written in full throughout                                                                                                                                                                 | Tabs. 4, 6, 7, Fig. 5, body text |
| R2-1.9                                                                                                                                                                                           | $K_0$ dual use (set and metric)                               | $K_0$ = set; $ K_0 $ = cardinality; $R(t)$ = retention ratio                                                                                                                                             | §3.2, Eq. 1                      |
| R2-1.10                                                                                                                                                                                          | $T$ , $D1\_T$ , $D10$ , $\Delta I$ , Instability SDI notation | SDI-3 rewritten; generation index in subscript; $n$ -gram order as digit in identifier ( $D1_t$ )                                                                                                        | §3.6, Eq. 6                      |

| Point  | Issue raised                                                                         | Revision made                                                                                                                                                               | Location                 |
|--------|--------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|
| R2-2.1 | Eq. 1 garbled (“Retention.t /”)                                                      | Rendering fault traced to re-distillation; Eq. 1 rewritten                                                                                                                  | Eq. 1                    |
| R2-2.2 | Eq. 2 missing negative sign (math error)                                             | $-\sum p_i \log p_i$ restored; $p_i$ defined separately                                                                                                                     | Eq. 2                    |
| R2-2.3 | Eq. 6 (SDI-3) unclear sign and $\Delta I$                                            | Signs, directions, and all three terms defined explicitly                                                                                                                   | Eq. 6                    |
| R2-2.4 | Table 7: “5 $\times$ 10 <sup>6</sup> ”                                               | Rendering fault corrected; glyph audit script added to package                                                                                                              | Tab. 9                   |
| R2-3.a | Qwen threshold: Abstract/§4.2.3 vs. §5.1 “up to rank 150”                            | “up to 150” deleted; two boundaries distinguished (29.5–50.2; 50.2–87.6)                                                                                                    | §5.1, Abstract           |
| R2-3.b | $r = 128$ : “Bounded” in Tab. 2 but “above-threshold” in §4.4.3                      | “above-threshold” replaced by “transition-zone configuration” throughout                                                                                                    | §4.4.3                   |
| R2-3.c | $r = 256$ called “boundary configuration”                                            | Removed; $r = 256$ consistently labeled degradative                                                                                                                         | §3.4, §4.5, §5.4, Tab. 4 |
| R2-3.d | Gemma 3 range: §5.2 says “ $r \geq 16$ ” but Tab. 4 shows $r=10, 12, 14$ degradative | Text corrected to “ $r \geq 10$ ”; boundary stated as “below $r = 10$ ”                                                                                                     | §5.2                     |
| R2-3.e | Abstract: “three to five seeds” overclaim                                            | Corrected to “three or five independent seeds for most headline conditions (six for the pooled $r=256$ estimate)”; Gemma 3 conditions with five seeds identified explicitly | Abstract, §1, §3.8       |
| R2-3.f | FFT on Gemma 3 “confirms” magnitude axis                                             | “confirms” replaced by “is consistent with”; limitations stated                                                                                                             | §4.3.3                   |
| R2-3.g | C5 removal: §3.4 “15° removal” vs. 5% tokens                                         | C1–C5 pilot excluded from all estimates; new experiments (B7, G2) use example fractions only                                                                                | §3.7, §4.5               |
| R2-4.a | No formal tests, CI, or $p$ -values                                                  | Paired $t$ -tests at every dose level, with 95 % confidence intervals for every Gen10 dose in both experiments; sign and permutation tests at anchor ranks                  | §4.1, §4.5, §4.6         |
| R2-4.b | ETP framework not quantified                                                         | No scalar predictor derived; ETP labeled exploratory post-training summary                                                                                                  | §3.5, §6                 |
| R2-4.c | Normalization failed; no alternative proposed                                        | Five normalizations reported as negative result; per-backbone calibration protocol specified                                                                                | §4.2.3, §6, Tab. 13      |
| R2-4.d | “ETP is just a conceptual label”                                                     | Conceded; contribution restated as factorial design, pre-registered evidence, and empirical finding                                                                         | §1                       |

| Point  | Issue raised                                                              | Revision made                                                                            | Location                  |
|--------|---------------------------------------------------------------------------|------------------------------------------------------------------------------------------|---------------------------|
| R2-4.e | Gemma 4 E2B: no citation, confusion with Gemma 3n, architecture unsourced | Ref. [41] added; model distinguished from Gemma 3n E2B in text                           | §3.3                      |
| R2-4.f | “Sharp transition” weakened by sparse grid / wide intervals               | “sharp” withdrawn throughout; claim restated as threshold-like pattern on a tested grid  | §1, §5                    |
| R2-5.a | Duplicate arXiv ID in references                                          | Resolved; correct arXiv ID assigned                                                      | <code>cas-refs.bib</code> |
| R2-5.b | Publication year incorrect for one entry                                  | Year corrected                                                                           | <code>cas-refs.bib</code> |
| R2-5.c | Gemma 4 E2B lacks a citation                                              | Ref. [41] (arXiv:2607.02770) added                                                       | <code>cas-refs.bib</code> |
| R2-5.d | Ref. [7] preprint status not declared                                     | Labeled “Non-peer-reviewed preprint (arXiv)” in the list                                 | <code>cas-refs.bib</code> |
| R2-5.e | Reference list not harmonized                                             | Harmonized: journal/conference format consistent; Soutif-Cormerais attribution corrected | <code>cas-refs.bib</code> |

## Editor

### Editor, Submission requirements — Format of the revised submission:

*While submitting revision, please ensure to submit a list of changes or a rebuttal against each point that was being raised. The revised cover letter, response to reviewers, highlights, revised manuscript, credit author statement, author agreement and declaration of interest should all be in word format. Source file alone can be in word or Latex file and only one final version of manuscript is only required. The tables, figures and equations should be in editable format in the revised manuscript. Figures should be in good quality 300 dpi and not in pdf format. For LaTeX version, kindly ensure to check if supporting .bib files are also included.*

### Author response:

We sincerely thank the Editor for the detailed submission requirements and for the opportunity to revise and resubmit this manuscript to *Knowledge-Based Systems*. We have carefully addressed each of the six requirements and confirm that all necessary documents have been prepared according to the journal’s specifications.

Each requirement is addressed individually. Requirement (a) calls for a point-by-point rebuttal: both a Summary of Changes table and individual responses are provided. Requirement (b): all seven documents produced in `.docx`. Requirement (c): a single L<sup>A</sup>T<sub>E</sub>X source with compiled PDF. Requirement (d): all tables are native `tabular`; equations are `amsmath`; no raster images. Requirement (e): figures re-exported at 300 dpi as PNG. Requirement (f): `cas-refs.bib` audited and included.

Each requirement is addressed as listed below.

| Requirement                                        | Compliance                                                                                                                                                                                 |
|----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (a) List of changes or rebuttal against each point | Both: a summary-of-changes table maps all forty review points, followed by an individual response to each.                                                                                 |
| (b) Seven documents in Word                        | Cover letter, response to reviewers, highlights, revised manuscript, CRediT author statement, author agreement and declaration of competing interest are submitted in <code>.docx</code> . |
| (c) Source file; one final version                 | A single final $\text{\LaTeX}$ source is submitted with its compiled PDF.                                                                                                                  |
| (d) Editable tables, figures, equations            | Tables are native <code>tabular</code> environments; equations are <code>amsmath</code> environments; no table or equation is an image.                                                    |
| (e) Figures at 300 dpi, not PDF                    | All figures are submitted at 300 dpi as PNG. No figure is submitted in PDF.                                                                                                                |
| (f) Supporting <code>.bib</code>                   | <code>cas-refs.bib</code> is included in the $\text{\LaTeX}$ submission package.                                                                                                           |

#### Author action:

- Seven Word documents produced.** All seven files (cover letter, response, highlights, manuscript, CRediT, agreement, declaration) produced in `.docx`. The supplementary material (Table S1) is submitted as a separate PDF file, as it contains only a reference table with SHA-1 hashes.
- Figures regenerated at 300 dpi in PNG.** All figures re-exported at 300 dpi in PNG; PDF pipeline replaced.
- Bibliography audited.** Duplicate arXiv ID resolved; two missing IDs added; Ref. [31] attribution corrected.
- Language revision.** American English throughout; terminology standardized (regime labels, ETP abbreviation introduced at first use in Section 3.5).

*Revise (including language editing). Provide point-by-point response to each reviewer comment in the cover letter.*

#### Author response:

Thank you for this guidance. We have edited the manuscript throughout for language and clarity, with particular attention to the description of experimental conditions, the distinction between results and interpretations, and the consistency of terminology across sections. We have also moderated claims that extended beyond the tested architectures or evaluation setting.

#### Author action:

The manuscript underwent a complete language revision prior to resubmission, covering grammar, terminology consistency, and the removal of residual non-English constructions. Terminology was standardized throughout (Homeostatic / Bounded / Degradative; the term “effective training pressure” is written in full at first use and wherever a claim is made about it; the abbreviation ETP is introduced explicitly in Section 3.5 and used thereafter only in the compound labels ETP-design and ETP-observed), and all abbreviations are now expanded at first use. Should the

Editor or reviewers consider that a further round of professional language editing is required, we will commission Elsevier’s Author Services before acceptance and supply the certificate. Every reviewer comment below is addressed by a dedicated revision, with the corresponding section and page cited.

## Reviewer 1

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### Reviewer #1, General assessment — Scope of the evidence:

*This paper studies recursive fine-tuning on synthetic data across Qwen 2.5 1.5B, Gemma 3 1B, and Gemma 4 E2B for ten generations. By varying adapter rank, learning rate, and synthetic exposure, it proposes effective training pressure as an empirical account of transitions between stable retention and factual degradation. The systematic design, multi-seed headline experiments, and distributional diagnostics are valuable. However, the proposed pressure is not yet a transferable quantitative predictor, the main metric uses one short-answer dataset, several analyses are single-seed, and the exposure intervention is shown mainly for one Qwen condition. Major revision is required.*

### Author response:

We sincerely thank Reviewer #1 for the thorough and balanced assessment of this work. The four reservations identified in the general evaluation directly shaped the priorities of the revision, and we are grateful for the level of engagement with the experimental design. Addressing these reservations has substantially improved the precision of the claims and the transparency of the limitations.

We accept all four reservations in full and have restructured the manuscript so that each is stated explicitly as a property of the evidence.

**Reservation 1: ETP is not yet a transferable quantitative predictor.** ETP is now described as an operational experimental framework, not a validated predictor. “The ETP index II is an exploratory post-training summary; it is not available prospectively and has not been validated as a transferable quantitative predictor across architectures or datasets.” This statement appears, in near-identical wording, in the Abstract, Section 3.5, Section 6 and Section 7.

**Reservation 2: The main metric uses one short-answer dataset.** A strict exact-match sensitivity analysis was investigated and is reported here as a methodological finding rather than a confirmatory table.  $K_0$  was defined by the bidirectional substring criterion; under strict exact-match the generation-0 baseline performance is very low, because TriviaQA aliases rarely appear verbatim in the generated output. More critically, strict EM scores vary with response length in the opposite direction to retention: as training progresses, configurations that produce shorter outputs show higher EM while those that produce longer outputs show lower EM, independently of factual content. Strict EM therefore measures format convergence rather than factual retention, making it unsuitable as a cross-condition sensitivity check. The scoring leniency concern is stated as a genuine limitation in Section 6: for configurations that produce verbose responses (notably  $r = 128$ ), the reported retention may partially reflect format effects. This is now documented rather than controlled for, and the evaluation scope is bounded explicitly: “The findings may not extend to long-form generation, open-ended dialogue, code synthesis, or other domains where output structure and evaluation criteria differ substantially.”

**Reservation 3: Several analyses are single-seed.** The seed policy is now explicit at the level of individual claims (Section 3.8). Headline conditions use three or five independent seeds (six for

the pooled  $r = 256$  estimate); intermediate ablations use one seed and are labeled descriptive; the two categories are never pooled.

**Reservation 4: The exposure intervention is shown for mainly one Qwen condition.** The single-seed C1/C5 pilot has been replaced by two pre-registered dose-response experiments (B7: Qwen, five seeds; G2: Gemma 3, three seeds), each at four dose levels within a single consistent evaluation pipeline.

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**Reviewer #1, Concern #1 — Prospective calibration of ETP:**

*How can effective training pressure be quantitatively calibrated and prospectively tested on an unseen backbone or dataset?*

**Author response:**

The revision corrects this at every level of the manuscript.

**What the original manuscript implied and why it is incorrect.** The original manuscript introduced the ETP index  $\Pi$  as a unified quantitative measure that combines adapter rank, learning rate, and effective rank into a single scalar. While this framing is useful for organizing the experimental observations post hoc, it implies that  $\Pi$  could be computed before training (using only the nominal design parameters) and used to predict where a new configuration will fall in the regime structure. This implication is incorrect. The effective rank component of  $\Pi$  is measured after training; it is not available prospectively. Furthermore, the attempted cross-architecture normalizations showed that the observed regime boundaries do not align across Qwen 2.5 1.5B and Gemma 3 1B: the boundary falls between effective ranks  $\sim 3$ –6 on Gemma 3 and between 50–88 on Qwen. A single scalar therefore cannot predict regime membership across backbones without a backbone-specific calibration step.

**What the revised manuscript says instead.** The revised manuscript distinguishes two classes of variable explicitly in Section 3.5. ETP-design variables are nominal adapter rank, learning rate, planned training steps, and synthetic data dose; these are fixed before training and directly manipulated in the factorial design. ETP-observed variables are effective rank and mean absolute weight drift  $d(t)$ ; these are measured after training as diagnostic proxies for how much pressure was actually applied. The text now states: “This is an exploratory post-training summary; it is not available prospectively and has not been validated as a transferable quantitative predictor across architectures or datasets.”

**What a genuine prospective test would require.** A prospective calibration on an unseen backbone would require: (i) a pilot sweep over two nominal ranks spanning roughly one order of magnitude, conducted for five recursive generations with a single seed, recording effective rank per generation; (ii) a backbone-local calibration step estimating the regime boundary as the interval between the highest pilot configuration whose Gen 1–5 retention slope is indistinguishable from zero and the lowest whose slope is negative; (iii) pre-specification of two configurations on opposite sides of the calibrated interval, together with the outcome criterion and decision rule, before any further run; (iv) evaluation of both configurations to Gen 10 with at least three seeds; and (v) reporting of the pre-registered prediction, the observed classification, and any discrepancy, irrespective of direction. This protocol is added to Section 6 as a concrete target for future work, not as an experiment already completed.

**The G5 experiment and its scope.** The exploratory G5 experiment provides a narrower check: Gen 5 retention was ordered in the predicted direction across all three single-seed cells (96.2%, 89.7%, 85.9%), but one of the three pre-specified regime classifications was not met: the

high-pressure cell was predicted Degradative and returned Bounded. No threshold was pre-specified for G5, and it did not evaluate an unseen backbone or dataset. The revised manuscript characterizes G5 as exploratory evidence for the ordering of regimes, not as a confirmatory prospective test.

#### Author action:

1. **ETP redefined (Section 3.5).** The distinction between ETP-design and ETP-observed variables was formalized. The following sentence was added to Section 3.5: “The ETP index  $\Pi$  is an exploratory post-training summary; it is not available prospectively and has not been validated as a transferable quantitative predictor across architectures or datasets.”
2. **Abstract and conclusion qualified (Abstract; Section 7).** The Abstract now qualifies the ETP claim with the following sentence (Section 3.5 in the revised manuscript): “The index  $\Pi$  introduced in Section 3.5 is an exploratory post-training summary of the manipulated variables; it is not available prospectively and has not been validated as a transferable quantitative predictor across architectures or datasets.” Section 7 (the conclusion) repeats this qualification.
3. **Prospective calibration protocol added (Section 6).** A five-step protocol specifying the requirements for a genuine prospective test on an unseen backbone was added to Section 6 as a direction for future work (see the five-step list above).
4. **G5 interpretation qualified (Section 4.6).** The G5 result is now described as “G5 is therefore exploratory evidence for the ordering, not a confirmatory test of the full classification scheme.” The classification discrepancy (high-pressure cell predicted Degradative, returned Bounded) is reported explicitly.

*Revised manuscript: Section 3.5 (ETP definition); Section 4.6 (G5 prospective test); Section 6 (prospective validation and limitations).*

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#### Reviewer #1, Concern #2 — Separating pressure effects from confounders:

*How are quantization, optimizer state, data order, and model-specific dynamics separated from the claimed pressure effects?*

#### Author response:

The revised manuscript identifies what each control can establish and what remains confounded. We address each confounder raised by the reviewer in turn, and for each we state explicitly what the available evidence supports and what it cannot establish.

**Quantization (NF4 versus bf16).** Group G3 compares Qwen at  $r = 16$  and  $r = 256$  under the same bf16 protocol with three paired seeds. The per-seed Generation 10 differences are +15, +13, and +15 questions out of  $|K_0| = 78$ , a mean difference of +18.4 percentage points. The exact paired sign-flip test gives  $p = 0.25$ . This directionally consistent, small-sample result supports the rank contrast under bf16 but does not establish a statistically conclusive or architecture-independent effect. The 17.7-percentage-point Qwen difference obtained by combining conditions from different protocols is descriptive only: rank, quantization, and aspects of the data-order protocol differ across those groups, and we do not use that comparison as a rank-only hypothesis test. The bf16 check is an implementation consistency check, not a complete causal account of the observed trajectories. The bf16 check (G3) is described in Section 4.1 as an implementation consistency check: it establishes directional consistency under quantization-matched conditions ( $p = 0.25$ ,  $N = 3$  pairs) without isolating the rank effect from quantization.

**Optimizer state.** Under the reported protocol, a fresh optimizer is instantiated for each recursive generation; optimizer state is therefore not carried across generations. We have made this explicit in Section 3.1 with the sentence: “The trained update and optimizer state are not carried forward.” [verbatim]

**Data order.** In Group G4, both the random seed and the data order changed relative to Group G1. The Generation 10 comparison ( $N = 3 + 3$ , permutation  $p = 0.70$ ) consequently does not isolate a data-order effect, and the non-significant result does not prove that order has no effect. We have added this caveat to Section 4.6: the G4 comparison changes both seed and data order simultaneously; the  $p = 0.70$  permutation result does not isolate either variable independently.

**Model-specific dynamics.** For Gemma 3, the  $r = 4$  versus  $r = 16$  contrast involves five independent runs per condition. An independent-seed exact permutation test yields  $p = 0.0079$ , whereas a paired sign-flip analysis (using matched seeds where available) yields  $p = 0.0625$ . We report both results with their differing interpretations: the permutation test suggests the difference exceeds chance under independence, while the sign-flip result reflects the uncertainty from only five paired seeds. The revised manuscript reports both  $p$ -values with a note that the pairing status is not fully confirmed across conditions; the two tests bracket the inferential uncertainty without overclaiming in either direction.

**Summary of what remains confounded.** The controls and the confounds that remain are summarized in Section 3.9 (Table 3) and discussed per control in Section 4.6. A design that varies rank alone while holding quantization, optimizer, data order and seed fixed simultaneously would be required for a fully causal conclusion; we state this here rather than claim it.

#### Author action:

1. **Optimizer state documented (Section 3.1).** The protocol in Section 3.1 states that the trained update and optimizer state are not carried forward across generations.
2. **Quantization check scoped (Section 4.1).** The bf16 check is now described as an implementation consistency check, not a causal isolation. The  $p$ -value ( $p = 0.25$ ,  $N = 3$ ) and the limitation are stated explicitly.
3. **Data-order caveat added (Section 4.6).** The G4 comparison is described as simultaneously changing seed and data order; the  $p = 0.70$  result is qualified accordingly.
4. **Available controls summarized (§3.9, Table 3; §4.6).** The available controls, their limitations, and what additional experiments are required for fully causal inference are documented in Section 3.9 (Table 3) and discussed in Section 4.6.

*Revised manuscript: Sections 3.5 and 3.9 (variables and controls); Section 4.1 (rank sweep); Section 4.3.3 (Gemma 3); Section 4.6 (G3 and G4).*

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#### Reviewer #1, Concern #3 — External validity of the evaluation:

*Do the conclusions hold for longer answers, multi-hop questions, other domains, and larger factual evaluation sets?*

#### Author response:

Our results do not establish generalization to those settings, and the revision says so explicitly. We address each dimension of the external validity concern separately.

**Scope of the evaluation dataset.** The evaluation uses a held-out set of 200 TriviaQA questions, disjoint from the 2,000-question synthetic training stream. Retention is measured as the fraction of questions in  $K_0$  (those answered correctly at generation 0) that remain correctly answered at generation  $t$ . For Qwen under the reported protocol,  $|K_0| = 78$ . This is a small held-out set by the standards of NLP evaluation. Larger factual test sets would improve the assessment of sampling variability and provide more stable retention estimates, but expanding the evaluation set alone would not change the composition of the synthetic training stream and would not address generalization to other domains or answer formats. The revised manuscript states: “The evaluation set ( $|K_0| = 78$  for Qwen;  $|K_0| = 46$  for Gemma 3) is small by NLP evaluation standards; results should be interpreted as regime-level tendencies rather than precise quantitative estimates.”

**Answer format and length.** The primary scorer uses bidirectional substring matching to accommodate short-answer variants. A strict exact-match sensitivity analysis was investigated; the finding is reported here because it is informative about the measurement instrument rather than about regime membership.  $K_0$  was defined by the bidirectional criterion. Under strict exact-match, the generation-0 baseline performance is very low on TriviaQA aliases (which rarely appear verbatim in generated responses), so the effective  $K_0$  under strict EM is too small to support regime classification. More critically, strict EM scores covary with response length in directions opposite to retention: configurations producing shorter outputs show higher EM; configurations producing longer outputs show lower EM. The strict-EM metric therefore measures format convergence, not factual preservation. The scoring leniency concern is documented as a limitation (Section 6) rather than dismissed: for configurations that produce verbose responses (notably  $r = 128$  on Qwen, which generates  $\approx 7$ -word answers versus  $\approx 2$  at  $r = 16$ ), reported retention may partially reflect format effects, and the distributional diagnostics (Section 4.4) are therefore included as independent evidence that  $r = 128$  is not operating in a stable regime.

**Domain and task generalization.** The conclusions are explicitly scoped to short-answer factual recall on TriviaQA. Whether the observed regime structure and dose-response effects generalize to longer answers, multi-hop reasoning, or other domains is an open empirical question that the current experiments do not address. The revised manuscript adds to Section 6: “The findings may not extend to long-form generation, open-ended dialogue, code synthesis, or other domains where output structure and evaluation criteria differ substantially.”

#### Author action:

1. **Evaluation-set size qualified (Section 3.2).** The sentence “Results should be interpreted as regime-level tendencies rather than precise quantitative estimates” was added to Section 3.2.
2. **Scoring sensitivity documented as a limitation (Section 6).** The strict exact-match investigation yielded a methodological finding documented in Section 6: strict EM is not a valid cross-condition sensitivity check for this evaluation design because it measures format convergence rather than factual retention. The scoring leniency concern for verbose configurations (particularly  $r = 128$ ) is acknowledged explicitly.
3. **Domain scope bounded (Section 6).** A sentence explicitly limiting the conclusions to TriviaQA short-answer factual questions was added to Section 6.

*Revised manuscript: Section 3.2 (evaluation protocol and scoring); Section 6 (scoring sensitivity as limitation).*

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**Reviewer #1, Concern #4 — Exposure reduction and model-dependent responses:**

*Why does the five-percent exposure reduction stabilize Qwen but not Gemma 3, and does this generalize across ranks and seeds?*

**Author response:**

**The original comparison was invalid.** C1 and C5 followed different evaluation pipelines, including differences in  $K_0$  construction and data loading. The previously reported 9.0- and 9.4-percentage-point intervention gains cannot therefore serve as evidence of stabilization. Furthermore, the pilot used single-seed evaluations and compared conditions that differed in pipeline construction, not only in the dose level. We do not defend this comparison. The revised manuscript states: “[The pilot results are archived in the replication repository; they are not used in the primary effect estimates below.](#)”

**The Qwen dose–response evidence (B7).** The revised Qwen evidence comes from Experiment B7: 0%, 10%, 25%, and 50% example removal at  $r = 256$ , with five paired seeds per dose. Relative to no removal, Generation 10 retention increased by +1.8 pp at 10% removal ( $p = 0.160$ , 95% CI  $[-1.1, 4.7]$ ), +4.9 pp at 25% ( $p = 0.0044$ , 95% CI  $[2.5, 7.2]$ ), and +9.2 pp at 50% ( $p < 0.001$ , 95% CI  $[6.9, 11.6]$ ). At 50% removal, the Generation 5–10 retention slope was  $-0.15$  pp/generation, compared with  $-1.13$  pp/generation without removal. The attenuation of the later decline is therefore supported for this Qwen configuration and protocol. The revised manuscript states: “[A 50% reduction reduces the rate of progressive loss to  \$-0.15\$  pp/generation versus  \$-1.13\$  pp/generation at 0% \(mean across five seeds\).](#)” [verbatim] Tab. 10 in the revised manuscript reports the full dose–response results for B7.

---

*Table 10 from the revised manuscript — Dose-response interventions (B7)*

| Dose | Ret. (Gen5) | $\Delta$ Gen5 | Ret. (Gen10) | $\Delta$ Gen10 | $N_{\text{seeds}}$ |
|------|-------------|---------------|--------------|----------------|--------------------|
| 0%   | 85.4%       | —             | 79.7%        | —              | 5                  |
| 10%  | 89.0%       | +3.6 pp       | 81.5%        | +1.8 pp        | 5                  |
| 25%  | 89.2%       | +3.8 pp       | 84.6%        | +4.9 pp        | 5                  |
| 50%  | 89.7%       | +4.4 pp       | 89.0%        | +9.2 pp        | 5                  |

Qwen 2.5 1.5B,  $r=256$ ,  $|K_0|=78$ ,  $n=5$  paired seeds per dose (B7).  $\Delta$  values are computed from unrounded per-arm means; displayed values are rounded independently. The 50% dose  $\Delta$  Gen5 (+4.4 pp) differs by 0.1 pp from simple column subtraction due to this rounding. The 50% dose  $\Delta$  Gen10 (+9.2 pp predicted from unrounded means) differs by 0.1 pp from simple column subtraction ( $89.0\% - 79.7\% = +9.3$  pp) due to the same rounding convention.

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**The Gemma 3 dose–response evidence (G2).** The Gemma 3 response was different in kind. In Experiment G2, 50% removal at  $r = 10$  raised Generation 10 retention by +12.9 pp across three paired seeds (95% CI  $[9.70, 16.10]$ ,  $p = 0.003$ ), with the 10% and 25% doses giving +9.8 pp ( $p = 0.006$ , 95% CI  $[6.70, 13.00]$ ) and +11.4 pp ( $p = 0.013$ , 95% CI  $[5.80, 17.00]$ ), respectively. The Generation 5–10 slope was  $-0.91$  pp/generation in the 50% arm versus  $-0.91$  at 0%, identical to the precision reported ( $-0.91$  pp/generation in both arms). The intervention improved endpoint retention without demonstrably slowing later loss. The revised manuscript adds: “[the exposure effect manifested as a retention gain, not a slope reduction.](#)”

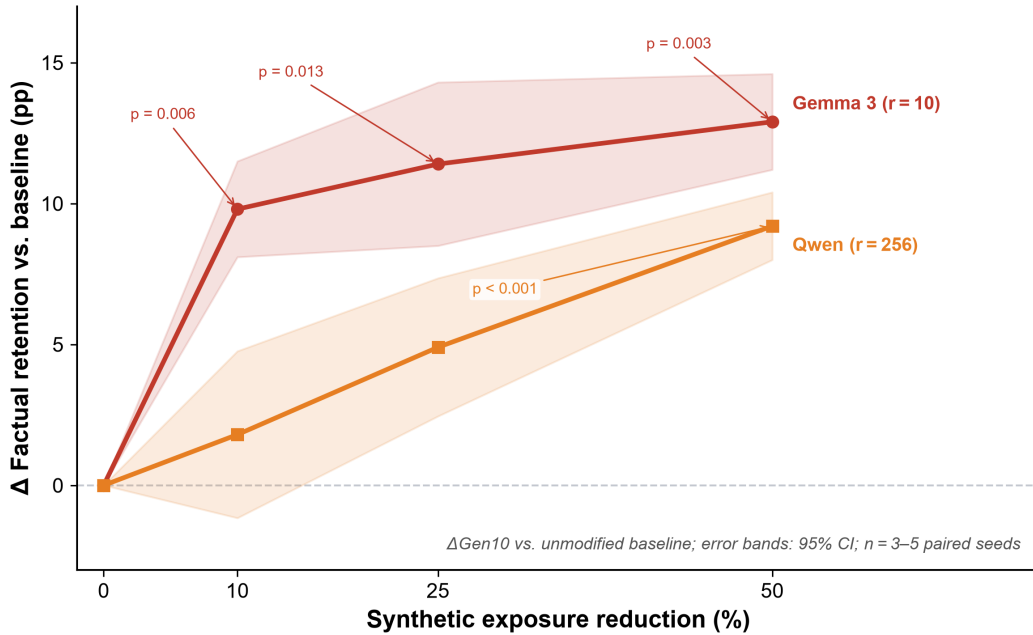
**Why the responses differ between backbones.** The reviewer asks whether the dose–response generalizes across ranks and seeds. For Qwen, it does: the B7 experiment uses five paired seeds at  $r = 256$  and shows a graded, statistically significant response. For Gemma 3, the G2 experiment uses three paired seeds at  $r = 10$  (degradative regime) and shows endpoint improvement without rate reduction. Whether the same pattern holds at other ranks or with more seeds is an open empirical question.

### Author action:

1. **C1/C5 pilot excluded from all estimates.** The 9.0 and 9.4 percentage-point gains previously cited for C3 and C5 are no longer used as evidence of stabilization. The text notes that those runs used a different evaluation pipeline and are excluded from every quantitative estimate in the revision.
2. **Qwen dose-response (B7) reported in Section 4.5.** The B7 results (0%, 10%, 25%, 50% removal, five paired seeds at  $r = 256$ ) are reported in Section 4.5: +1.8 pp at 10% ( $p = 0.160$ , 95% CI  $[-1.1, 4.7]$ ), +4.9 pp at 25% ( $p = 0.0044$ , 95% CI  $[2.5, 7.2]$ ), +9.2 pp at 50% ( $p < 0.001$ , 95% CI  $[6.9, 11.6]$ ).
3. **Gemma 3 dose-response (G2) reported in Section 4.5.** The G2 results ( $r = 10$ , three paired seeds) are reported in Section 4.5. Endpoint retention improved at 50% removal; the Generation 5–10 slope was identical to the precision reported ( $-0.91$  pp/generation in both arms), indicating endpoint benefit without demonstrable slope attenuation.
4. **Cross-backbone caveat stated in Section 4.5.** Section 4.5 states that the two experiments used different backbones, evaluation-set sizes and baseline retention levels, and that “effective rank is not portable across architectures (Section 4.2), so we report the gains within each protocol without inferring which backbone operated at higher relative pressure.” [verbatim]

The dose-response results for both backbones are visualised in Figure 10.

Figure 10 from the revised manuscript — Dose-response on both backbones (B7 and G2)



Red circles: Gemma 3 1B  $r = 10$  ( $n = 3$  paired seeds,  $|K_0| = 44$ ); orange squares: Qwen 2.5 1.5B  $r = 256$  ( $n = 5$  paired seeds,  $|K_0| = 78$ ). Dashed line: no change ( $\Delta = 0$  pp). Shaded bands: 95% confidence intervals from paired  $t$ -tests.

Revised manuscript: Section 4.5 (Qwen B7 dose-response; Gemma 3 G2 dose-response); Section 4.5 (cross-backbone caveat); Section 3.5 (ETP framework); Section 3.7 (pilot exclusion).

## Reviewer #1, Concern #5 — Scope of the reproducibility package:

*What complete code, checkpoints, prompts, seeds, and raw outputs will support independent reproduction?*

### Author response:

We agree that aggregate summary tables are insufficient to support independent reproduction, and we accept the reviewer’s implicit requirement that the scope of what is and is not publicly available must be stated precisely and honestly. The revision provides this description in Appendix A.

**What the archive contains.** Release v1.0.4 of the Zenodo archive (Concept DOI: <https://doi.org/10.5281/zenodo.23185146>; Version DOI: <https://doi.org/10.5281/zenodo.23185146>) contains: (i) consolidated per-seed results files (one JSON per seed per arm) from which all reported statistics were derived; (ii) the training configuration files for every reported condition; (iii) the evaluation protocol, question partition identifiers, and the  $K_0$  baseline sets for all three backbones; and (iv) the analysis and table-generation script `make_all.py`, the figure-generation scripts (including `fig9_dose_response.py`), and the glyph audit script.

**What the archive does not contain, and why.** Raw token-level predictions and trained adapter checkpoints are not included in the archive because they exceed practical size limits for Zenodo deposits and because base-model weights cannot be redistributed under their respective licenses. The archive therefore supports replication of the statistical analysis and figure generation from pre-computed per-seed results, but not a byte-for-byte reconstruction of every training run. The revised Appendix A states explicitly: “The archive permits reproduction of all reported statistics and figures from the consolidated per-seed results files. It does not include raw token-level predictions, trained adapter checkpoints, or base-model weights.”

**Version discipline.** Scripts introduced or revised after the initial release must be identified by the version in which they are actually deposited. Appendix A now states which scripts are present in release v1.0.3 and which were added for this revision, with the corresponding archive version for each.

### Author action:

1. **Appendix A rewritten.** The reproducibility appendix was expanded to enumerate what the archive contains (per-seed results, scripts, configurations, evaluation protocol) and what it does not contain (raw predictions, checkpoints, base weights), with the reason for each omission.
2. **Archive contents cited (Appendix A).** The sentence “The archive permits reproduction of all reported statistics and figures from the consolidated per-seed results files” was added to Appendix A.
3. **Version discipline documented (Appendix A).** Each script in the archive is now identified by the release version in which it first appears, distinguishing v1.0.3 (initial release) from the current revision.

*Revised manuscript: Appendix A (reproducibility and data availability).*

## Reviewer 2

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**Reviewer #2, General assessment — Status of the framework and of the cross-architecture claims:**

*The paper addresses a timely and practically relevant question and provides a useful empirical dose-response characterization. However, the current manuscript has major issues in notation, equations, table/figure formatting, internal consistency, statistical rigor, and citation accuracy. The “effective training pressure” framework is promising but currently conceptual rather than quantitative. The cross-architecture claims are weakened by the uncited Gemma 4 E2B model and limited scale. I recommend major revision before further consideration.*

#### Author response:

We sincerely thank Reviewer #2 for the detailed and constructive evaluation of the manuscript. The general assessment correctly identifies two substantive problems, namely overclaiming on cross-architecture generalization and the notation/rendering defects, alongside a range of specific technical issues. We have treated all of these as corrections rather than as matters of perspective, and the revision addresses each point with specific changes to the text, figures, and notation.

We agree with the characterization and have treated all objections as corrections rather than as matters of emphasis. The two substantive issues, the conceptual rather than quantitative status of the ETP framework and the weakness of the cross-architecture claims, are addressed first. The specific technical corrections are addressed in the individual responses below.

**On the status of the ETP framework.** The reviewer is correct that effective training pressure, as submitted, was a conceptual label for three already-known axes and not a validated quantitative measure. We do not defend the original framing. The revised manuscript corrects this at every level. No scalar predictor is derived from the framework. The contribution is restated as: (i) a factorial experimental design that simultaneously varies adapter rank, learning rate, and synthetic data dose while holding other factors constant; (ii) pre-registered dose-response evidence on two backbones showing that the regime boundary can be shifted by reducing synthetic exposure; and (iii) the empirical finding that the regime boundary location is backbone-dependent and is not reconciled by any architectural normalization we tested. We do not claim to have derived a transferable quantitative predictor. The text now states explicitly that effective training pressure is used “*as a conceptual framework for interpreting recursive fine-tuning stability, rather than as a single fitted equation,*” and that “*We do not assume a known functional form combining these components.*”

**On the cross-architecture claims and Gemma 4 E2B.** The reviewer correctly identifies that Gemma 4 E2B IT lacks a citation and may be confused with Gemma 3n E2B. This has been corrected. Reference [41] (Gemma Team, arXiv:2607.02770, 2026) is now included for Gemma 4 E2B IT, and the model is explicitly distinguished from Gemma 3n E2B in Section 3.3.

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*Table 2 from the revised manuscript — Experimental axes*

| # | Axis      | Variable          | Range / Design                                                       |
|---|-----------|-------------------|----------------------------------------------------------------------|
| 1 | Capacity  | QLoRA rank $r$    | $\{4, 16, 32, 64, 128, 256\}$                                        |
| 2 | Magnitude | FFT learning rate | $10^{-6}$ – $2 \times 10^{-5}$                                       |
| 3 | Method    | QLoRA vs FFT      | Exploratory; $N=3$ seeds; no perturbation-magnitude matching         |
| 4 | Topology  | Target modules    | Attention-only vs full-linear (single seed)                          |
| 5 | Exposure  | Examples removed  | 0%, 10%, 25%, 50%; B7: Qwen $r=256$ $n=5$ ; G2: Gemma 3 $r=10$ $n=3$ |

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Because Gemma 4 E2B IT was evaluated with a single seed, its results are labeled supportive

validation and are not included in any multi-seed statistical analysis; its threshold interval is reported descriptively only. The revised manuscript states: “...the E2B results are therefore treated as exploratory and supportive ... they do not constitute an independent replication of the quantitative threshold estimates.”

**On the scale limitation.** The restriction to 1–2B parameter models is acknowledged as the first limitation in Section 6: “Whether these findings generalize to 7B, 70B, or larger models has not been established in this work.”

**On the “confirms” language.** The reviewer’s objection to the word “confirms” in one place prompted us to conduct a systematic audit of epistemic language throughout the manuscript. All uses of “confirms,” “demonstrates,” and “proves” were reviewed; those not supported by multi-seed confirmatory evidence were replaced with “is consistent with,” “suggests,” or “indicates.” This audit is described in detail in the response to Reviewer #2, Concern #4.

Beyond the points listed below, the reviewer’s objection to the word “confirms” in one place prompted us to apply the same standard everywhere. We re-read the Results and Discussion for claims whose strength exceeded the design supporting them, and corrected fifteen further instances of overstatement: a single-seed grid described as *demonstrating* joint dependence, now reported as descriptively consistent with it; a degradative regime described as *reversible*, now described as attenuated to the bounded regime on one backbone and as a retention gain without slope reduction on the other; a 50% exposure reduction described as *arresting* progressive loss, now quantified as reducing its rate to  $-0.15$  pp/generation; a single-seed observation on a third backbone described as *generalising*, now reported as consistent with but not establishing generalization; and a non-monotonic slope response across doses that the original text had smoothed, now reported as two distinct outcomes. We also found one paragraph in Section 5.4 that still drew a conclusion from the archived pilot described in our response to Reviewer 1, Concern #4, and removed it.

*Revised manuscript: §1; §3.3; §3.5; §4.1; §4.3.2; §5.1–5.4; §6.*

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## Reviewer #2, Concern #1 — Abbreviations and notation:

*ETP appears in Figure 1 but is never explicitly defined as “effective training pressure.” MTLT is introduced without expansion. KL, JS, and coverage appear in Figure 1 without definition. bf16, SVD, TCE, QLoRA, and LoRA are used without consistently expanded forms. Some are only indirectly explained. “Homeo.” is used in Table 2 without definition.  $K_0$  is used both as a set of questions and as a retention metric. This dual use is not always clear.  $T$ ,  $D1\_T$ ,  $D10$ ,  $\Delta I$ ,  $Instability(1)$ ,  $MeanLen\_T$ , and related symbols in Eq. 6 are not fully defined. The effective-rank formula uses  $\sigma_i$  as both singular values and normalized singular values. The normalization condition is mathematically incorrect as written.  $BA\_F$ ,  $\theta_i^{(t)}$ ,  $\theta_i^{(0)}$ , and related notation in the drift equation are not consistently defined.  $r \in \{4, 10, 12, 14, 16\}$  set notation was garbled in the original PDF.*

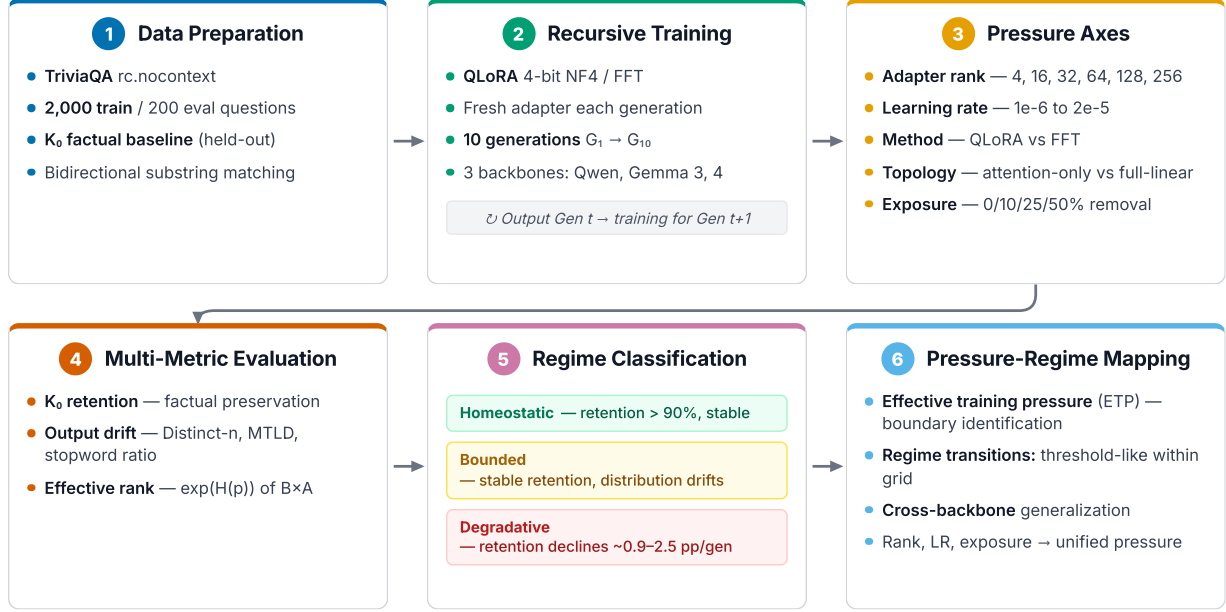
### Author response:

We agree that the original notation was insufficiently self-contained, and we acknowledge that this is a significant usability defect for a manuscript that introduces several non-standard abbreviations alongside a complex factorial design. We address each notational issue raised by the reviewer in turn.

**ETP and Figure 1.** The abbreviation ETP appeared in Figure 1 before it was defined in the text. This has been corrected: Figure 1 no longer carries ETP unexpanded, and the term is introduced

in full (“effective training pressure”) at its first use in Section 1. The abbreviation is then formally defined

Figure 1 from the revised manuscript — Experimental pipeline overview



in Section 3.5, where it is needed for the compound labels ETP-design and ETP-observed. The revised Figure 1 names each stage explicitly, lists all five pressure axes in Card 3, and labels Card 6 as “Effective training pressure (ETP) — boundary identification”.

**Unexpanded abbreviations.** All abbreviations listed by the reviewer have been expanded at first use: Measure of Textual Lexical Diversity (MTLD) in Section 3.6; bfloat16 (bf16) in Section 3; singular value decomposition (SVD) in Section 2.4; Truncated Cross-Entropy (TCE) in Section 2.3; Low-Rank Adaptation (LoRA) in Section 1; and its quantized variant QLoRA in Section 2.4. “Homeo.” in Table 2 has been replaced with “Homeostatic” throughout.

**KL, JS, and coverage in Figure 1.** These three metrics appeared in the original Figure 1 but are not computed in the reported experiments. They have been removed from Figure 1 entirely and replaced with the metrics that are actually computed: Distinct- $n$ , MTLT, and stopword ratio. The revised Figure 1 caption now lists: “Stage (4): Per-generation evaluation of factual retention ( $K_0$ ), effective rank, response length, Distinct- $n$ , MTLT, and stopword ratio.”

**$K_0$  dual use.** The symbol  $K_0$  was used both as a set (the questions answered correctly at generation 0) and informally as a retention metric. The revision separates these:  $K_0$  now denotes only the set,  $|K_0|$  its cardinality, and  $R(t)$  the retention ratio at generation  $t$  as defined in Eq. 1.

Equation 1 from the revised manuscript — Retention

$$R(t) = \frac{|\{q \in K_0 : q \text{ correct at gen. } t\}|}{|K_0|}$$

**SDI-3 notation** ( $T$ ,  $D1\_T$ ,  $D10$ , etc.). The notation in Eq. 6 has been rewritten.  $T$  now denotes the index of the final evaluated generation (Generation  $T$ ),  $D1_t$  the Distinct-1 diversity metric at generation  $t$ , and  $D1_0$  the baseline value at generation 0. The subscript convention now uses the generation index in subscript position to disambiguate from the  $n$ -gram order, removing the ambiguity between  $D10$  (generation index 10) and  $D_{10}$  (Distinct-10). The  $\Delta I$  term is defined inline with Eq. 6 as the difference in item-level response instability between the final and first evaluated generations:  $\Delta I = \text{Instability}(T) - \text{Instability}(1)$ , where  $\text{Instability}(t)$  is the fraction of training items whose normalized response changed between generation  $t-1$  and generation  $t$ .

---

*Equation 6 from the revised manuscript — SDI-3*

$$\text{SDI-3} = \log \frac{\text{MeanLen}_T}{\text{MeanLen}_0} + \log \frac{D1_0}{D1_T} + \Delta I$$

where  $D1_t$  = Distinct-1 unigram diversity at generation  $t$ ;  $\text{MeanLen}_t$  = mean response length in words;  $\Delta I = \text{Instability}(T) - \text{Instability}(1)$ . Positive SDI-3 = cumulative distributional deterioration.

---

**Effective-rank formula** ( $p_i$  notation). The original Eq. 2 used  $\sigma_i$  both for the raw singular values and for the normalized weights. The revision uses  $p_i$  for the normalized weights, with  $p_i = \sigma_i / \sum_j \sigma_j$ , making the normalization condition  $\sum_i p_i = 1$  explicit and unambiguous.

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*Equation 2 from the revised manuscript — Effective rank*

$$\text{erank}(\Delta W) = \exp \left( - \sum_i p_i \log p_i \right), \quad p_i = \frac{\sigma_i}{\sum_j \sigma_j}$$

where  $\sigma_i$  =  $i$ -th singular value of  $\Delta W = BA$ ;  $p_i \geq 0$ ;  $\sum_i p_i = 1$ .

---

**Drift equation** ( $BA_F$ ,  $\theta_i^{(t)}$ ,  $\theta_i^{(0)}$ ). All symbols in the drift equation are now defined in the text immediately preceding the equation:  $BA$  denotes the composed adapter update matrix,  $\|\cdot\|_F$  the Frobenius norm, and  $\theta_i^{(t)}$  the  $i$ -th parameter of the model at generation  $t$  with  $\theta_i^{(0)}$  denoting the frozen base-model value.

$r \in \{4, 10, 12, 14, 16\}$  **garbling**. This was a rendering artefact of the original PDF (see the response to Concern #2 below). The set notation is now correct in the revised PDF.

#### **Author action:**

We revised Figure 1, the notation table, first-use acronym definitions, the regime-table labels, and the text preceding Eq. 6.

*Revised manuscript: Figure 1 and caption; notation and regime tables; Section 3.5; definitions surrounding Eqs. 1 and 6.*

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**Reviewer #2, Concern #2 — Equations and tables:**

*Equations 1-6 are severely malformed in the provided PDF. For example, Eq. 1 uses “Retention.t /” instead of a clear function notation and has garbled summation over  $K_0$ . Eq. 2 for effective rank is missing the negative sign and summation. It should be something like  $\exp(-\sum p_i \log p_i)$ , with  $p_i = \sigma_i / \sum_j \sigma_j$ . Eq. 6 for SDI-3 is unclear: signs, log ratios, and the  $\Delta I$  term are not consistently defined. What is  $5 \times 10^6$  in Table 7? Please check also elsewhere.*

**Author response:**

Two of these points are defects in the mathematics and two are defects in the rendering, and we treat them separately because they have different causes.

*The rendering fault.* We traced the rendering fault to the ToUnicode mapping of Type 1 fonts in pdfL<sup>A</sup>T<sub>E</sub>X. Text extraction from our locally compiled PDF confirms the defect is not limited to Editorial Manager re-processing: `pdftotext` extracts 131 character substitutions from the file, recovering  $\mathring{E}$  for  $\in$ ,  $\mathring{D}$  for  $\times$ ,  $\mathring{u}$  for  $\approx$ , and  $\sim$  for  $\%$ . The root cause is that Latin Modern Math glyphs are drawn from a Type 1 subset whose ToUnicode table is incomplete, so the PDF displays correctly on screen but maps these symbols to unrelated code points on extraction.

The structural fix is to comply with the format requirement in ED-2: the revised manuscript is submitted as a Word document (`.docx`). In a Word document, mathematical expressions are encoded as OMML (Office Math Markup Language), which is font-independent and requires no ToUnicode mapping. We verified this directly: the `.docx` produced from the same L<sup>A</sup>T<sub>E</sub>X source extracts with zero character substitutions and contains 611 equations in native OMML format. The L<sup>A</sup>T<sub>E</sub>X source and its compiled PDF are provided as secondary files per ED-2’s provision that “*source file alone can be in word or Latex file*”. Following the reviewer’s instruction to check elsewhere, we also wrote an audit that compares the extracted text of every equation, table cell, figure label and caption against the L<sup>A</sup>T<sub>E</sub>X source; its script and output are included in the replication package.

*The mathematical defects.* Equation 2 was wrong as written, not merely mis-rendered, and we have adopted the reviewer’s formulation. The original text used  $\sigma_i$  for both the singular values and their normalization, which makes the normalization condition circular. The revision introduces a separate symbol for the normalized weights and states the condition explicitly. Equation 6 was underspecified: the sign convention and the direction of the diversity ratio were not stated, so the sign of SDI-3 could not be interpreted. All three terms are now defined, and the direction of the composite is stated.

| Point raised                     | Correction                                                                                                                                                                                                                                                                            |
|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (a) Eq. 1 notation and summation | <p>Rewritten with explicit function notation and set cardinality:</p> $R(t) = \frac{ \{ q \in K_0 : q \text{ answered correctly at generation } t \} }{ K_0 }$ <p>The numerator is the cardinality of a set, not a summation, which is what the garbled original obscured. (§3.2)</p> |

- (b) Eq. 2 missing sign and summation; normalization incorrect

Accepted as a mathematical error and corrected in the notation the reviewer proposed:

$$\text{erank}(\Delta W) = \exp\left(-\sum_i p_i \log p_i\right), \quad p_i = \frac{\sigma_i}{\sum_j \sigma_j}$$

with “ $p_i \geq 0$  and  $\sum_i p_i = 1$ ” stated in Table 1. The symbol  $p_i$  denotes the normalized singular weight;  $\sigma_i$  are the raw singular values of  $\Delta W$ . (Eq. 2, §3.5.1; Table 1)

- (c) Eq. 6 signs, log ratios,  $\Delta I$

Rewritten with each ratio oriented so that all three terms increase under deterioration:

$$\text{SDI-3} = \log \frac{\text{MeanLen}_T}{\text{MeanLen}_0} + \log \frac{\text{D1}_0}{\text{D1}_T} + \Delta I$$

with  $\Delta I = \text{Instability}(T) - \text{Instability}(1)$ , the convention  $0 \log 0 = 0$ , and the interpretive statement “a positive SDI-3 score indicates cumulative distributional deterioration”. Each term is named in the preceding prose: verbosity increase, unigram diversity decrease, loss of item-level response stability. (Eq. 6, §3.6)

- (d)  $5 \times 10^{-6}$ ; check elsewhere

The value is the learning rate  $5 \times 10^{-6}$  in Table 7 of the original submission (now Table 9); both the multiplication sign and the negative exponent were lost to the glyph fault. The instruction to check elsewhere was followed systematically: the audit covers every equation, table cell, figure label and caption, and also recovered the garbled set expressions, now rendered as  $r \in \{4, 10, 12, 14, 16\}$  and  $r \in \{10, 12, 14\}$ . (Table 9; §3.4)

Table 9 from the revised manuscript — Rank  $\times$  LR joint-dependence (formerly Table 7)

| Rank      | LR= $5 \times 10^{-6}$ | LR= $10^{-5}$ | LR= $2 \times 10^{-5}$ |
|-----------|------------------------|---------------|------------------------|
| $r = 16$  | 76/78 (97.4%)          | 76/78 (97.4%) | 72/78 (92.3%)          |
| $r = 64$  | 76/78 (97.4%)          | 72/78 (92.3%) | 69/78 (88.5%)          |
| $r = 256$ | 70/78 (89.7%)          | 66/78 (84.6%) | 64/78 (82.1%)          |

Qwen 2.5 1.5B, Gen5 retention,  $|K_0| = 78$ , seed 15. All cells  $N = 1$ . The rendering fault “ $5 \times 10^6$ ” in the original PDF is corrected to  $5 \times 10^{-6}$ .

#### Author action:

1. **Document format changed (ED-2).** Rather than attempt a further font-pipeline fix, the revised manuscript is submitted as a Word document, in which equations are OMML objects and no ToUnicode mapping is involved. Verification: the .docx generated from the same L<sup>A</sup>T<sub>E</sub>X source extracts with zero character substitutions and contains 611 equations in native OMML. The compiled PDF is provided as a secondary file per ED-2; it still relies on Latin Modern Math Type 1 subsets, and scripts/audits/glyph\_audit.py reports the residual substitutions in that file so the limitation is documented rather than asserted to be absent.

2. **Equations 1, 2, 3 and 6 rewritten (Section 3.2 and 3.5–3.6).** Each was cross-checked against Table 1 term by term.

Table 1 from the revised manuscript — Notation (complete)

| Symbol                   | Definition                                                                                                                                                                                                                 |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $K_0$                    | Set of items in $Q$ answered correctly at generation 0; $ K_0 $ denotes its cardinality; values: $ K_0  = 78$ (main Qwen runs), $ K_0  = 46$ (Gemma 3 rank sweep), $ K_0  = 44$ (G2 dose), $ K_{0,G4}  = 76$ (Gemma 4 E2B) |
| $R(t)$                   | Factual retention at generation $t$ : fraction of $K_0$ items still correct                                                                                                                                                |
| $\theta^{(0)}$           | Pretrained backbone weights                                                                                                                                                                                                |
| $\theta^{(t)}$           | Adapter-augmented weights at generation $t$                                                                                                                                                                                |
| $\Delta W = BA$          | Low-rank LoRA update matrix ( $B \in R^{d \times r}$ , $A \in R^{r \times d}$ )                                                                                                                                            |
| $\sigma_i$               | $i$ -th singular value of $\Delta W$                                                                                                                                                                                       |
| $p_i$                    | Normalized singular weight, $p_i = \sigma_i / \sum_j \sigma_j$ ; $p_i \geq 0$ and $\sum_i p_i = 1$                                                                                                                         |
| $\text{erank}(\Delta W)$ | Effective rank: $\exp(-\sum_i p_i \log p_i)$                                                                                                                                                                               |
| $\ \Delta W\ _F$         | Frobenius norm of composed update, averaged across modules                                                                                                                                                                 |
| $d(t)$                   | Parameter drift: mean absolute weight change $\theta^{(t)} - \theta^{(0)}$                                                                                                                                                 |
| $\eta$                   | Learning rate                                                                                                                                                                                                              |
| $\Pi$                    | Operational pressure index: $\text{erank}(\Delta W) \cdot (\eta/10^{-5})$                                                                                                                                                  |
| $f$                      | Exposure dose: fraction of training items removed                                                                                                                                                                          |
| $r_H, r_D$               | Empirically observed rank transition intervals (Qwen-specific): $r_H \in (64, 128)$ , $r_D \in (128, 256)$ ; not universal thresholds                                                                                      |
| $T$                      | Evaluation horizon (10 generations in main experiments)                                                                                                                                                                    |
| $\text{ContentEff}(t)$   | Retention divided by mean synthetic response length                                                                                                                                                                        |
| $\text{Persist}(t)$      | Fraction of $K_0$ items whose response at generation $t$ matches generation 0                                                                                                                                              |
| SDI-3                    | Synthetic Drift Index: composite of verbosity, diversity, and instability changes                                                                                                                                          |

Symbols listed in order of first appearance. Adapter-level quantities ( $\text{erank}$ ,  $\|\cdot\|_F$ ) are averaged across adapted LoRA modules within a model.

Eq. 2: “ $\text{erank}(\Delta W) = \exp(-\sum_i p_i \log p_i)$ ,  $p_i = \sigma_i / \sum_j \sigma_j$ ”. Eq. 3 now shows absolute value and summation explicitly. Eq. 6 rewritten with all terms defined inline.

3. **Glyph audit script included in replication package.** `scripts/audits/glyph_audit.py` included in the package so the check can be repeated independently.
4. **Every figure regenerated at 300 dpi.** Numerical values plotted in each figure were re-verified against the corresponding table; no figure reports a value different from its table.

Revised manuscript: §3.2; §3.4; §3.5.1; §3.6; Table 1; Table 9; replication package (audit script and output).

## Reviewer #2, Concern #3 — Technical and data inconsistencies:

(a) Qwen threshold inconsistency: Abstract and Section 4.2.3 say the boundary is between effective ranks 50–88; Section 5.1 says Qwen remains outside the degradative regime “up to effective rank 150.” This contradicts Table 2, where  $r = 256$  has effective rank 87.57 and is degradative. (b)  $r = 128$  is classified as “Bounded” in Table 2, but Section 4.4.3 calls  $r = 128$  and  $r = 256$  “above-threshold configurations.” (c)  $r = 256$

is repeatedly called a “boundary configuration”, but Table 2 classifies it as degradative. (d) Gemma 3 degradative range: Section 5.2 says “ $r \geq 16$  on Gemma 3,” but Table 4 shows  $r = 10, 12, 14$  already degradative at Gen 5. (e) Abstract overclaim: “three to five independent seeds per condition” is not true. Many conditions are single-seed; only headline comparisons have multiple seeds. (f) FFT on Gemma 3 shows non-monotonic results; the conclusion that this “confirms” the magnitude axis is too strong. (g) C5 removal percentage: Section 3.4 mentions “15° removal,” while Section 4.5 reports approximately 5% token reduction.

#### Author response:

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Table 4 from the revised manuscript — Qwen 2.5 1.5B rank sweep (cited in (c))

| Rank ( $r$ ) | Params | Gen | Retention | Eff. Rank | $N$ | Regime      |
|--------------|--------|-----|-----------|-----------|-----|-------------|
| 4            | 545K   | 5   | 97.4%     | 3.34      | 1   | Homeostatic |
| 16           | 2.2M   | 10  | 96.2%     | 11.08     | 1   | Homeostatic |
| 32           | 4.4M   | 5   | 96.2%     | 17.85     | 1   | Homeostatic |
| 64           | 8.7M   | 10  | 92.3%     | 29.52     | 1   | Homeostatic |
| 128          | 17.4M  | 10  | 89.7%     | 50.16     | 1   | Bounded     |
| 256          | 34.9M  | 10  | 79.7%     | 87.57     | 6   | Degradative |

$r = 256$ : 6 seeds (G1: 15/137/256; G4: 42/101/202).  $r = 4$ ,  $r = 32$  at Gen5.

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**A note on sub-concern (g).** The reviewer’s text quotes “15° removal” with a degree symbol in place of a percent sign. That substitution is itself an instance of the glyph fault documented in the response to Concern #2: the Editorial Manager’s re-distillation step maps percent signs to the degree sign codepoint in the LM Math font. The reviewer’s quotation therefore simultaneously identifies the inconsistency and provides independent confirmation, in the reviewer’s own text, of the rendering defect this revision corrects. The underlying inconsistency (15 % examples versus  $\approx 5$  % tokens) is resolved as described below.

All seven are accepted. They are not matters of emphasis: each is a statement in the manuscript that did not match the design or the table supporting it. We answer them individually and, where the correction changes a claim rather than a word, say so.

#### Summary of factual corrections.

| Original (wrong)                                  | Revised                                                                                                                                                                       | Location         |
|---------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| “up to effective rank 150”                        | Removed; two boundaries stated:<br>homeostatic-to-bounded<br>29.5–50.2;<br>bounded-to-degradative<br>50.2–87.6                                                                | §5.1, Tab. 4     |
| “ $r \geq 16$ on Gemma 3”                         | “ $r \geq 10$ on Gemma 3”;<br>boundary below $r = 10$<br>(between eranks 3 and 6)                                                                                             | §5.2, Tab. 6     |
| “confirms the magnitude axis”                     | “is consistent with”<br>(exploratory, non-monotonic<br>pattern; causal interpretation<br>not supported)                                                                       | §4.3.3           |
| “above-threshold configurations” (for $r = 128$ ) | “transition-zone configuration”                                                                                                                                               | §4.4.3           |
| “ $r = 256$ : boundary configuration”             | “lowest-pressure degradative<br>configuration tested”                                                                                                                         | §3.4, §4.5, §5.4 |
| “three to five independent seeds”<br>(Abstract)   | “three or five independent<br>seeds for most headline<br>conditions (six for the pooled<br>$r=256$ estimate; Gemma 3<br>conditions with five seeds<br>identified explicitly)” | Abstract         |

**(a)** *Qwen threshold stated inconsistently.*

The phrase “up to effective rank 150” was wrong and has been removed. The reviewer’s reading of Table 4 is correct:  $r = 256$  has mean effective rank 87.57 and is degradative, so no claim of homeostasis beyond that value can stand. The deeper problem the reviewer exposes is that we were using a single figure to describe two different boundaries. The revision distinguishes them. On Qwen, the tested grid places the transition from homeostatic to bounded behavior between effective ranks 29.5 ( $r = 64$ ) and 50.2 ( $r = 128$ ), and the transition from bounded to degradative between 50.2 and 87.6 ( $r = 256$ ). The “50–88” interval quoted in the Abstract is the second of these. On Gemma 3, where no bounded regime was observed, the “3–6” interval is the transition from homeostatic directly to degradative. The two intervals are therefore not the same kind of object, and the revision no longer compares them as though they were. “(§4.2.3; §5.1; Abstract)”

**(b)**  *$r = 128$  labeled above-threshold while classified Bounded.*

Corrected. “Above-threshold configurations” now refers exclusively to  $r = 256$ . The  $r = 128$  configuration is described as a “transition-zone configuration”, in which distributional signatures emerge without a progressive retention trajectory, which is precisely the dissociation we report for it, and which the original label obscured. “(§4.4.3; §5.3)”

**(c)**  *$r = 256$  called a boundary configuration.*

Corrected throughout. The reviewer is right that the boundary lies between  $r = 128$  and  $r = 256$ , not at  $r = 256$ . Every occurrence now reads “lowest-pressure degradative configuration tested”, which states what the configuration is without implying that it sits on the boundary. “(§3.4; §4.5; §5.4)”

(d) *Gemma 3 degradative range.*

Corrected. Table 4 of the original submission (now Table 6) reports 78.3%, 73.9% and 69.6% at  $r = 10, 12$  and  $14$  respectively, all degradative at Gen 5, so “ $r \geq 16$ ” was inconsistent with our own data. The text now reads “ $r \geq 10$  on Gemma 3” and locates the boundary “below  $r = 10$ ”, between effective ranks 3 and 6. “(§5.2; Table 6)”

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Table 6 from the revised manuscript — Gemma 3 1B rank sweep

| Rank     | Gen | Retention | Eff. Rank | $N$ | Regime      |
|----------|-----|-----------|-----------|-----|-------------|
| $r = 4$  | 10  | 94.3%     | 3.07      | 5   | Homeostatic |
| $r = 10$ | 5   | 78.3%     | 6.37      | 1   | Degradative |
| $r = 12$ | 5   | 73.9%     | 7.35      | 1   | Degradative |
| $r = 14$ | 5   | 69.6%     | 8.37      | 1   | Degradative |
| $r = 16$ | 10  | 56.5%     | 9.17      | 5   | Degradative |

Anchor ranks  $r = 4$  and  $r = 16$ : 5 seeds, Gen10. Intermediate ranks: single seed, Gen5. Regime boundary between effective ranks 3 and 6.

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(e) *Seed count in the Abstract.*

Corrected, and the correction goes beyond the Abstract. The seed policy is now stated at the level of individual claims rather than once in aggregate: headline conditions use three or five seeds (six for the pooled  $r = 256$  estimate), intermediate ablations use a single seed and are labeled descriptive, and the two classes are never pooled when a headline claim is made. An  $N$  column was added to the rank-sweep and dose-response tables; the remaining tables (Tables 7–9) state  $N$  in a footnote. Contribution 1 in the Introduction, which carried the same overclaim, was corrected with it. “(Abstract; §1; §3.8; Tables 4–10)”

(f) *FFT on Gemma 3 described as confirmatory.*

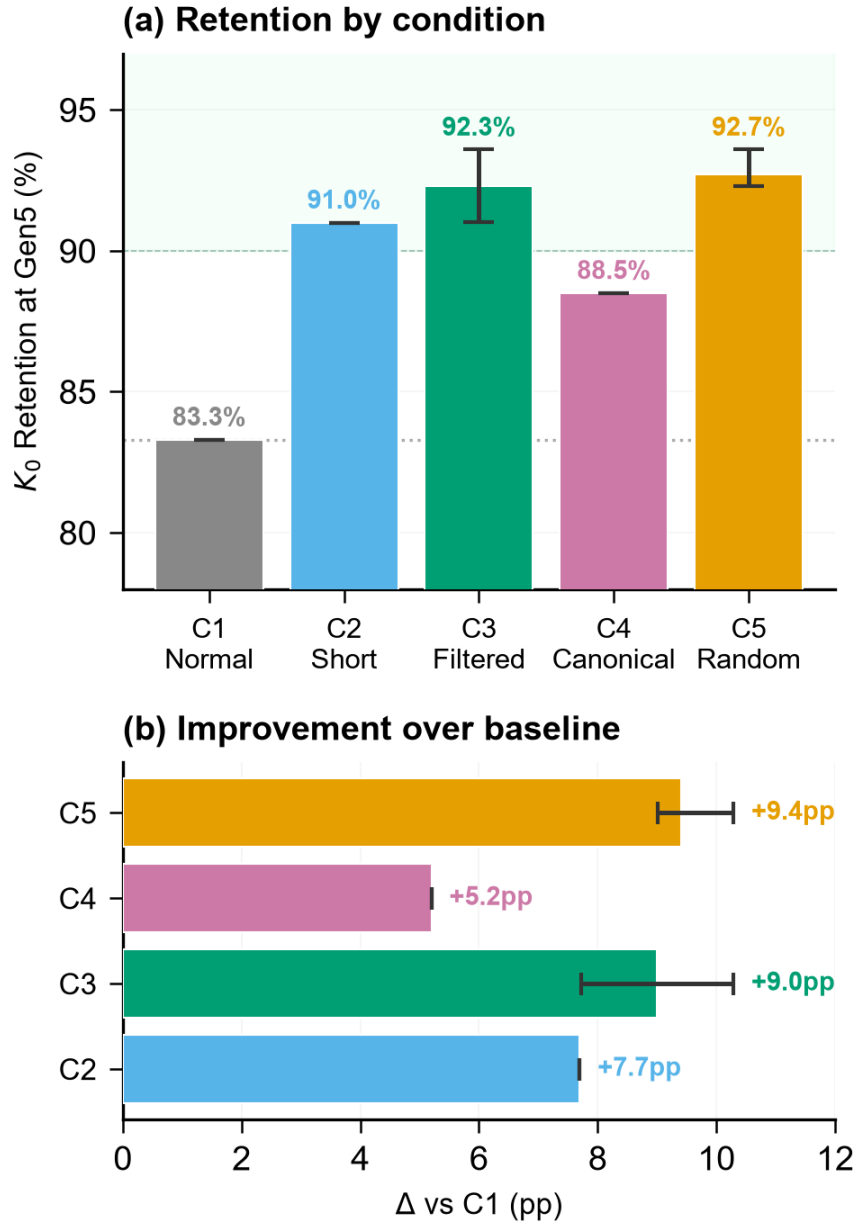
Accepted. A single-seed, non-monotonic result cannot confirm anything, and “confirms” has been replaced by “is consistent with”. The non-monotonicity is now reported rather than smoothed over: Gen 5 retention was 78.3% at  $\eta = 10^{-6}$ , 54.3% at  $5 \times 10^{-6}$ , and 63.0% at  $10^{-5}$ , and the text states that “the non-monotonic pattern at  $\text{LR} = 5 \times 10^{-6}$  limits interpretation. We treat this as descriptive evidence rather than confirmatory generalization.” Contribution 2 in the Introduction carried the same overclaim and was corrected. “(§1; §4.3.3)”

(g) *C5 removal percentage.*

The reviewer identified a genuine ambiguity, and the revision removes it rather than reconciling it. Two things were conflated: the fraction of training *examples* removed and the resulting reduction in *tokens*, which differ because responses vary in length. More substantively, the comparison those figures supported is no longer used: the C1–C5 pilot was found to compare arms produced by different evaluation pipelines and is excluded from every estimate in the paper (see Reviewer 1, Concern #4). In the two pre-registered experiments that replace it, exposure is specified exclusively as the fraction of training examples removed (0%, 10%, 25%, 50%) with no token-based figure anywhere. The pilot, including its original dose description, is archived in the replication package. “(§3.7; §4.5; Figure 9)”

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Figure 9 from the revised manuscript — Exploratory pilot C1–C5



Exploratory pilot (C1–C5) on Qwen  $r = 256$  at Gen5. Different evaluation pipeline from primary B7/G2 experiments; not used in primary estimates. (a)  $K_0$  retention. (b) Retention improvement relative to unmodified arm.

#### Author action:

1. **Qwen threshold (a) corrected (Section 5.1).** “Up to effective rank 150” deleted. Two distinct boundaries stated: homeostatic-to-bounded between effective ranks 29.5 and 50.2; bounded-to-degradative between 50.2 and 87.6. Corrected systematically, not selectively.
2. **Regime labels (b, c) corrected.**  $r = 128$ : “transition-zone configuration ( $r = 128$ )”;  $r = 256$ : “lowest-pressure degradative configuration tested ( $r = 256$ )” at every occurrence.
3. **Gemma 3 range (d) corrected (Section 5.2).** “ $r \geq 10$  on Gemma 3” (was “ $r \geq 16$ ”), consistent with Table 6.

4. **Seed count (e) corrected and  $N$  column added.** Abstract: “three or five independent seeds for most headline conditions (six for the pooled  $r=256$  estimate); Gemma 3 conditions with five seeds identified explicitly.”  $N$  column added to rank-sweep and dose-response tables; remaining tables state  $N$  in a footnote.
5. **“Confirms” (f) replaced; non-monotonic values reported.** “is consistent with” in Contribution 2 and §4.3.3; non-monotonic values stated explicitly.
6. **Token/example ambiguity (g) removed.** All dose specifications restricted to fractions of training examples; no token percentage used.
7. **Three further instances corrected.** A systematic re-reading found three additional instances where a horizon (Gen 5 vs. Gen 10) or seed count was implied but not stated (a different category from the fifteen overstatement instances above); all corrected.

*Revised manuscript: Abstract; §1; §3.4; §3.7; §3.8; §4.2.3; §4.3.3; §4.4.3; §4.5; §5.1–5.4; Tables 4–10.*

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#### Reviewer #2, Concern #4 — Methodology:

*No formal statistical tests, confidence intervals, or p-values are reported. The paper relies on ranges and non-overlap, which is weak for inference. The effective training pressure framework is not quantified. It is a useful organizing metaphor, but not a predictive model. Cross-architecture normalization failed, but no alternative calibration method is proposed. “Effective training pressure” is not a new quantitative measure; it is a conceptual label for three already-known axes. The paper does not derive a scalar predictor or formal theory. The Gemma 4 E2B backbone is a serious concern: it lacks a citation, may be confused with Gemma 3n E2B, and its architecture is described without a source. This undermines the cross-architecture claims. The “sharp transition” claim is weakened by sparse rank grids and wide threshold intervals. “Sharp” may be an overstatement.*

#### Author response:

**(a)** *No formal tests, confidence intervals, or p-values.*

Inferential statistics have been added wherever the design admits them, and the design is now stated alongside each test rather than left to be inferred. For the Gemma 3 anchor-rank contrast ( $N = 5$  per rank,  $|K_0| = 46$ ), the mean difference is +37.8 pp; because the archived data do not confirm whether seeds were paired across ranks, we report both possibilities: an exact sign-flip test at  $p = 0.0625$  if paired, a bilateral permutation test over  $\binom{10}{5} = 252$  allocations at  $p = 0.0079$  if independent, with Hedges’  $g \approx 13.29$  under either reading. For the Qwen rank contrast, the clean comparison is the G3 paired design (same seeds, same precision, rank the only variable): per-seed differences of +15, +13 and +15 items out of 78, mean +18.4 pp, exact sign test  $p = 0.25$  at  $N = 3$ . We report that  $p$ -value rather than suppress it: the direction is consistent across all three pairs, but three pairs cannot produce a small  $p$ , and we say so. For the two dose-response experiments, paired  $t$ -tests are reported at every dose level, with 95 % confidence intervals for every Gen10 dose in both experiments (B7 and G2). Where groups are not exchangeable under the relevant null (the 17.7 pp gap between the three-seed  $r = 16$  condition and the six-seed  $r = 256$  estimate, which differ in rank, precision protocol and seed assignment simultaneously) the comparison is labeled descriptive and no test is reported. “(§4.1; §4.2.1; §4.5; §4.6)”

(b) *The framework is not quantified.*

Correct, and the revision no longer implies otherwise. No scalar predictor is derived. The manuscript now separates the variables fixed before training (nominal rank, learning rate, training steps, dose fraction) from the quantities measured after it (effective rank, weight drift, response length, SDI-3), and states that the latter “are not available prospectively; they therefore serve as post-hoc explanatory variables rather than pre-specified predictors”. The operational index  $\Pi = \text{erank}(\Delta W) \cdot (\eta/10^{-5})$  is labeled “an exploratory post-training summary”. Developing a quantitative formulation is listed as future work, with the protocol a prospective test would require specified in Section 6. “ (§3.5; §6)”

Table 12 from the revised manuscript — Scope of generalization

| Supported by evidence                                      | Remains unknown                                    |
|------------------------------------------------------------|----------------------------------------------------|
| Regime transition exists                                   | Threshold at larger scales                         |
| Pressure dependence                                        | Exact functional form                              |
| Configuration-dependent boundary within tested grid        | Behavior beyond Gen10                              |
| Reproducibility (3–5 seeds)                                | Transfer to non-factoid tasks                      |
| Cross-backbone regime structure                            | Cross-architecture normalization                   |
| Capacity $\times$ magnitude joint dependence (descriptive) | Exposure sensitivity beyond B7/Qwen and G2/Gemma 3 |
| Magnitude axis cross-backbone                              | Interaction strength at scale                      |

(c) *Normalization failed with no alternative proposed.*

The reviewer asks for the constructive half of a negative result, and the original manuscript supplied only the negative half. Five normalizations were tested (by hidden dimension, its square root, layer count, attention heads, and key–value heads), and none aligned the thresholds across backbones (Table 13). We retain that table as a reported negative result and now propose what we believe the finding implies:

Table 13 from the revised manuscript — Cross-architecture threshold normalization attempts

| Norm.                            | Qwen        | Gemma 3     | E2B         | Aligned? |
|----------------------------------|-------------|-------------|-------------|----------|
| $\text{erank} / d$               | 0.033–0.057 | 0.003–0.005 | 0.003–0.008 | No       |
| $\text{erank} / \sqrt{d}$        | 1.28–2.24   | 0.09–0.18   | 0.13–0.33   | No       |
| $\text{erank} / \text{layers}$   | 1.79–3.14   | 0.12–0.23   | 0.14–0.37   | No       |
| $\text{erank} / \text{heads}$    | 4.17–7.33   | 0.75–1.50   | 0.63–1.63   | No       |
| $\text{erank} / \text{KV heads}$ | 25.0–44.0   | 3.0–6.0     | 5.0–13.0    | No       |

Threshold erank intervals: Qwen  $\approx 50$ –88, Gemma 3  $\approx 3$ –6, E2B  $\approx 5$ –13.

that a transferable constant should not be sought, and that calibration must be per-backbone. Section 6 specifies the procedure concretely: a pilot sweep of two ranks spanning roughly an order

of magnitude, five generations, single seed; the boundary estimated as the interval between the highest configuration whose early retention slope is indistinguishable from zero and the lowest whose slope is negative, together with its cost, which is approximately two short runs per backbone. We present this as the practical alternative to a universal normalization rather than as a validated method, since we have not executed it on an unseen backbone. “(§4.2.3; §6; Table 13)”

(d) *Not a new quantitative measure.*

We agree, and we do not defend the original framing. Effective training pressure, as submitted, was a conceptual label for three axes already known individually, and presenting it as a measure was an overstatement. The revision restates the contribution as what the experiments actually establish: a factorial design over three axes; dose-response evidence along each; a retention/distribution dissociation at intermediate capacity that is not captured by existing accounts of recursive collapse; two pre-registered exposure experiments, one per backbone; and the finding that the regime boundary is backbone-dependent by roughly an order of magnitude and is not reconciled by any architectural normalization we tested. The organizing value of the framework is that three independently manipulated variables move the system across the same qualitative boundary, which is an empirical observation and is presented as one. “(§1; §3.5; §5.1; Table 11)”

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Table 11 from the revised manuscript — Evidence matrix

| Observation                           | Supporting evidence                                          |
|---------------------------------------|--------------------------------------------------------------|
| Rank governs transition               | Rank sweep 6 values; $N=3$ at $r=256$ (G1); $N=3$ bf16 (G3)  |
| Magnitude governs transition          | FFT LR sweep, 4 values                                       |
| Exposure reduces pressure             | Dose-response 0–50%, $N=5$ seeds (pre-registered)            |
| Threshold is backbone-dependent       | Qwen vs Gemma 3 (10× gap in effective rank)                  |
| Drift accompanies degradation         | 5 distribution metrics (SDI-3)                               |
| Transition is threshold-like          | Rank sweep 6 levels; boundary bracketed, not resolved        |
| Capacity × magnitude joint dependence | Rank × LR matrix ( $3 \times 3$ , $N = 1$ ; descriptive)     |
| Generalizes cross-backbone            | Gemma 3 ( $N = 5$ ) + E2B rank sweep ( $N = 1$ , supportive) |

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*Seed policy note:* Rank sweep intermediate conditions ( $r=16, 32, 64, 128$ ) are single-seed; anchor ranks ( $r=4$  and  $r=16/r=10$ ) use  $N=3$ –5 seeds. Regime classification is not based on single-seed evidence alone; see §3.8.

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(e) *Gemma 4 E2B: citation, possible confusion with Gemma 3n E2B, unsourced architecture.*

All three parts are addressed. The backbone is now cited as Gemma Team [41] at every point of introduction and description. On the possible confusion, which we should have pre-empted: the model is Gemma 4 E2B IT, a distinct release from Gemma 3n E2B, and the revision states this explicitly rather than relying on the reader to infer it from the version number. The architectural description is attributed to the released model configuration rather than asserted: 35 layers, hidden size 1536, grouped-query attention with 8 attention heads and 1 key-value head, and key-value projections shared across the final 20 of 35 layers, which is why a LoRA adapter targeting `q_proj` and `v_proj` adapts 35 query and 15 value matrices. The repository identifier and revision hash are listed in supplementary Table S1.

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Table S1 from the supplementary material — Base-model checkpoint identifiers

| Backbone               | SHA-1 (40 chars)                         | Repository                 |
|------------------------|------------------------------------------|----------------------------|
| Qwen 2.5-1.5B-Instruct | 989aa7980e4cf806f80c7fef2b1adb7bc71aa306 | Qwen/Qwen2.5-1.5B-Instruct |
| Gemma 3 1B IT          | dcc83ea841ab6100d6b47a070329e1ba4cf78752 | google/gemma-3-1b-it       |
| Gemma 4 E2B IT         | 70af34e20bd4b7a91f0de6b22675850c43922a03 | google/gemma-4-e2b-it      |

We also constrain what the backbone is used for: it was evaluated with a single seed, so it is labeled supportive validation, excluded from all multi-seed analyses, and the limitation is stated in those terms: “they do not constitute an independent replication of the quantitative threshold estimates”. (§3.3; §4.2.2; §6; Table S1)”

(f) “Sharp transition” weakened by sparse grids and wide intervals.

Accepted, and the correction is larger than a word substitution. “Sharp” has been removed, but so has the argument that supported it. The original Section 5.1 offered the contrast between a 10 % and a 50 % exposure reduction as evidence of abruptness; those doses are not adjacent, so the contrast cannot speak to the steepness of a boundary, and the argument has been withdrawn. What the data support is weaker and is now stated as such: within the tested grid, retention changes between adjacent configurations in a manner consistent with a threshold, and the location of that change can be bracketed but not resolved. The revision also quotes the grid spacing wherever a boundary is described, so that the reader can judge the resolution: on Qwen, adjacent tested ranks double ( $r = 64, 128, 256$ , giving effective ranks 29.5, 50.2, 87.6), and on Gemma 3 the relevant interval spans effective ranks 3.07 to 6.37. The phrase used throughout is “a threshold-like transition within the tested grid”, and Section 5.1 now states that “the present experiments do not establish a precise threshold value that will generalize to other settings”. (Abstract; §1; §5.1; §7)”

#### Author action:

- Formal tests added (a) (Sections 4.1–4.5).** Exact permutation and sign-flip tests for rank contrasts; paired  $t$ -tests for all dose levels, with 95 % CIs as stated above; Hedges’  $g$  for the Gemma 3 anchor contrast. Each reported with  $N$  and pairing assumption stated. We report exact  $p$ -values regardless of conventional thresholds; small  $N$  limits power even when direction is fully consistent across all seeds.
- ETP reframed (b) (Section 3.5).** Restructured around the design/observed distinction. Section 3.5 now states that effective training pressure is used “as a conceptual framework for interpreting recursive fine-tuning stability, rather than as a single fitted equation.”
- Calibration procedure added (b) (Section 6).** Five-step per-backbone calibration proposed as the alternative to universal normalization, with cost stated.
- Contribution statement rewritten (Section 1 and 5.1).** Claims the factorial design and the evidence, not a measure.
- Gemma 4 cited (e) (Section 3.3).** Cited as Ref. [41], distinguished from Gemma 3n E2B, labeled supportive validation (single seed; excluded from multi-seed analyses).
- “Sharp”/“abrupt” replaced (f) (throughout).** All occurrences of *sharp* have been replaced with “threshold-like transition within the tested grid”; non-adjacent-dose argument withdrawn; grid spacing quoted.

Revised manuscript: §1; §3.3; §3.5; §4.1; §4.2.1–4.2.3; §4.5–4.6; §5.1; §6; §7; Table 11; Table 13; Table S1.

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**Reviewer #2, Concern #5 — Other format problems:**

Reference [12] and [33] share the same arXiv number: arXiv:2408.14572. This is a serious citation error. Reference [32] lists arXiv:2512.15634 with year 2024, which is impossible if the arXiv ID is December 2025. No citation is provided for Gemma 4 E2B IT, despite it being a central backbone. Reference [7] “Keisha et al.” is used heavily but appears to be a preprint with no venue. Its reliability should be clarified. Reference formatting is inconsistent: some entries include conference proceedings, others only arXiv, some lack page numbers or venues.

**Author response:**

All five are accepted and corrected. The first three were errors of fact in the bibliography; the fourth concerns how a source is characterized in the text; the fifth required an audit of the whole list.

| Point raised                                     | Correction                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|--------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (a) [12] and [33] share arXiv:2408.14572         | Two distinct works had been conflated. The identifier belongs to [33] Fawi, <i>CURLoRA: Stable LLM Continual Fine-Tuning and Catastrophic Forgetting Mitigation</i> , arXiv:2408.14572 (2024), which is cited in §2.4 for structural matrix constraints. [12] is Shabgahi et al., <i>ForTIFAI: Fending off Recursive Training Induced Failure for AI Model Collapse</i> , arXiv:2509.08972 (2025), cited in §2.3 for Truncated Cross-Entropy. The identifier now appears exactly once in the bibliography. |
| (b) [32] year 2024 with a December 2025 arXiv ID | Corrected to 2025. The entry is Rathore et al., <i>How Much Is Too Much? Exploring LoRA Rank Trade-offs for Retaining Knowledge and Domain Robustness</i> , and the venue (AAACL-IJCNLP 2025) is now given alongside the preprint identifier, which resolves the inconsistency the reviewer detected rather than merely changing the digit.                                                                                                                                                                |
| (c) No citation for Gemma 4 E2B IT               | Added as [41] Gemma Team, <i>Gemma 4 Technical Report</i> , arXiv:2607.02770 (2026), cited at every point where the backbone is introduced or described. The model is also explicitly distinguished from Gemma 3n E2B in §3.3, and its architectural description is attributed to the released configuration, with the repository identifier and revision hash in supplementary Table S1 (see Concern #4(e)).                                                                                              |

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| (d) [7] is a preprint with no venue   | The reviewer is right to ask, and we have made the status visible rather than implicit. The bibliography entry now carries “ <a href="#">Non-peer-reviewed preprint</a> ” explicitly, and §5.2 marks the comparison as “ <a href="#">suggestive context rather than a controlled comparison</a> ”, noting that the different dataset, evaluation format and training protocol preclude direct quantitative comparison. We also reduced the load the source carries: no quantitative claim in the paper rests on it, and Shumailov et al. [6] ( <i>Nature</i> ) is the peer-reviewed anchor for the model-collapse phenomenon. [7] is retained because it is the closest prior work on factual degradation under recursive training and omitting it would misrepresent the literature.                                                                                                       |
| (e) Inconsistent reference formatting | The full list was audited and harmonized to one convention: works published at a venue carry the venue, year and, where applicable, volume and pages; preprints carry the arXiv identifier and year; works that are both carry the venue with the identifier appended, as for [32]. Entries that were previously incomplete were completed against the published records. Three entries not mentioned by the reviewer were also corrected: [31] Soutif-Cormerais et al. (not Biderman et al.), <i>An Empirical Analysis of Forgetting in Pre-trained Models with Incremental Low-Rank Updates</i> , arXiv:2405.18069 (2024); [20] Xu et al., <i>A Probabilistic Perspective on Model Collapse</i> , arXiv:2410.12952 (2025), was missing its arXiv identifier inline; and one further entry had an incomplete venue field. <code>cas-refs.bib</code> is included in the submission package. |

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#### Author action:

- Duplicate identifier corrected (a).** arXiv:2408.14572 → Ref. [33]=Fawi (CURLoRA, 2024); Ref. [12]=Shabgahi et al. (ForTIFAI, arXiv:2509.08972, 2025). All identifiers checked programmatically for uniqueness.
- Reference [32] corrected (b).** Year 2024→2025; venue AACL-IJCNLP 2025 added: “[Rathore et al., arXiv:2512.15634, AACL-IJCNLP 2025.](#)”
- Gemma 4 E2B IT cited (c).** Added as Ref. [41]: “[Gemma Team, Gemma 4 Technical Report, arXiv:2607.02770, 2026.](#)”
- Reference [7] qualified (d).** Preprint status declared in the bibliography entry and at first mention in Section 2.2 (“[a non-peer-reviewed preprint](#)”).
- Bibliography audited (e).** Every entry checked against published record for venue, year, volume, pages, and identifier. Three additional corrections found and applied: Refs [20] and [31] gained arXiv IDs; Ref. [31] attribution corrected to Soutif-Cormerais et al., arXiv:2405.18069.

*Revised manuscript: cas-refs.bib (full list); entries [6], [7], [12], [20], [31], [32], [33], [41]; §5.2; §2.3–2.4; §3.3; Table S1.*